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Basic Concepts of Lie Algebra

1.1 Definitions and first examples

Definition 1.1.1

A vector space L over a field $\mathbb{F}$, with an operation $L \times L \rightarrow L$, denoted $(x, y) \mapsto [xy]$ and called the **bracket** or **commutator** of x and y , is called a **Lie algebra** over $\mathbb{F}$ if the following axioms are satisfied:

- (L1) The bracket operation is bilinear.
- (L2) $[xx] = 0$ for all $x \in L$.
- (L3) $[x[yz]] + [y[zx]] + [z[xy]] = 0 \quad (x, y, z \in L)$.

Remark.

- (L1) and (L2), applied to $[x + y, x + y]$ imply anticommutativity :
(L2') $[xy] = [-yx]$.
- Conversely, if $\text{char } \mathbb{F} \neq 2$, it is clear that (L2') will imply (L2).

Definition 1.1.2

We define a **Lie subalgebra** of L to be a vector subspace $K \subseteq L$ such that

$$[x, y] \in K \quad \text{for all } x, y \in K.$$

Example 1.1.3

Suppose that V is a finite-dimensional vector space over F . Write $\mathfrak{gl}(V)$ for the set of all linear maps from V to V . This is again a vector space over F , and it becomes a Lie algebra, known as the **general linear algebra**, if we define the Lie bracket $[-, -]$ by

$$[x, y] := x \circ y - y \circ x \quad \text{for } x, y \in \mathfrak{gl}(V),$$

where $\circ$ denotes the composition of maps.

1.2 Lie algebras of derivations

Definition 1.2.1

- By an ***F*-algebra** (not necessarily associative) we simply mean a vector space $\mathfrak{A}$ over F endowed with a bilinear operation $\mathfrak{A} \times \mathfrak{A} \rightarrow \mathfrak{A}$, usually denoted by juxtaposition (unless $\mathfrak{A}$ is a Lie algebra, in which case we always use the bracket).
- By a ***derivation*** of $\mathfrak{A}$ we mean a linear map $\delta : \mathfrak{A} \rightarrow \mathfrak{A}$ satisfying the familiar product rule $\delta(ab) = a\delta(b) + \delta(a)b$ for each $a, b \in \mathfrak{A}$.

1.3 Ideals and homomorphism

1.3.1 Ideals

Definition 1.3.1

A subspace I of a Lie algebra L is called an ***ideal*** of L if $x \in L, y \in I$ together imply $[xy] \in I$.

Remark.

由于 $[x, y] = -[y, x]$, 理想不分左右.

1.3.2 Homomorphisms and representations

Definition 1.3.2

- A linear transformation $\phi : L \rightarrow L'$ (L, L' Lie algebras over F) is called a ***homomorphism*** if $\phi([xy]) = [\phi(x)\phi(y)]$, for all $x, y \in L$.
- ϕ is called a ***monomorphism*** if $\text{Ker}\phi = 0$, an ***epimorphism*** if $\text{Im}\phi = L'$, an ***isomorphism*** if it is both mono- and epi-.

Definition 1.3.3

If L_1 and L_2 are Lie algebras over a field F , then we say that a map $\varphi : L_1 \rightarrow L_2$ is a ***homomorphism*** if φ is a linear map and

$$\varphi([x, y]) = [\varphi(x), \varphi(y)] \quad \text{for all } x, y \in L_1.$$

Notice that in this equation the first Lie bracket is taken in L_1 and the

second Lie bracket is taken in L_2 . We say that φ is an *isomorphism* if φ is also bijective.

Example 1.3.4

An extremely important homomorphism is the *adjoint homomorphism*. If L is a Lie algebra, we define

$$\text{ad} : L \rightarrow \mathfrak{gl}(L)$$

by $(\text{ad } x)(y) := [x, y]$ for $x, y \in L$. It follows from the bilinearity of the Lie bracket that the map $\text{ad } x$ is linear for each $x \in L$. For the same reason, the map $x \mapsto \text{ad } x$ is itself linear. So to show that ad is a homomorphism, all we need to check is that

$$\text{ad}([x, y]) = \text{ad } x \circ \text{ad } y - \text{ad } y \circ \text{ad } x \quad \text{for all } x, y \in L;$$

this turns out to be equivalent to the Jacobi identity. The kernel of ad is the centre of L .

Proof. We have from Jacobi identity that

$$\begin{aligned} \text{ad}([x, y])(z) &= [[x, y], z] \\ &= [x, [y, z]] + [y, [z, x]] \\ &= [x, \text{ad } y(z)] - [y, \text{ad } x(z)] \\ &= \text{ad } x \circ \text{ad } y(z) - \text{ad } y \circ \text{ad } x(z) \end{aligned}$$

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